Birthday Guessing Game

This program guesses the user's birthday using five sets of numbers. The user is asked whether their birthday appears in each set. The program then calculates the birthday by adding the first number from each set in which the birthday appears. After displaying the result, the program automatically returns to the first set and repeats the process.

set 1 = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31
set 2 = 2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31
set 3 = 4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31
set 4 = 8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31
set 5 = 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31

Code implementation :
The program was written in Python using Visual Studio Code (VS Code) as the code editor

```
hile True:
    birthday = 0
    set_1 = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
    print(set_1)
    answer = int(input("Is your birthday included in this set? (Enter 1 for
Yes and 0 for No): "))
    if answer == 1:
        birthday = birthday + set_1[0]
    set_2 = [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
    print(set_2)
    answer = int(input("Is your birthday included in this set? (Enter 1 for
Yes and 0 for No): "))
    if answer == 1:
        birthday = birthday + set_2[0]
    set_3 = [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
    print(set_3)
    answer = int(input("Is your birthday included in this set? (Enter 1 for
Yes and 0 for No): "))
    if answer == 1:
        birthday = birthday + set_3[0]
    set_4 = [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
    print(set_4)
    answer = int(input("Is your birthday included in this set? (Enter 1 for
Yes and 0 for No): "))
```

```
    if answer == 1:
        birthday = birthday + set_4[0]
    set_5 = [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
    print(set_5)
    answer = int(input("Is your birthday included in this set? (Enter 1 for
Yes and 0 for No): "))
    if answer == 1:
        birthday = birthday + set_5[0]
    print("Your birthday is:", birthday)
```

Example output :

```
[1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
Is your birthday included in this set? (Enter 1 for Yes and 0 for No): 1
[2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
Is your birthday included in this set? (Enter 1 for Yes and 0 for No): 0
[4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
Is your birthday included in this set? (Enter 1 for Yes and 0 for No): 1
[8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set? (Enter 1 for Yes and 0 for No): 1
[16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set? (Enter 1 for Yes and 0 for No): 0
Your birthday is: 13
```

While loop allows program to automatically returning to the beginning

```
[1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
Is your birthday included in this set? (Enter 1 for Yes and 0 for No):
```